

Telco Customer Churn Analysis & Retention Strategy

Internship / Portfolio Project — Final Report

Telco Customer Churn — Executive Retention Dashboard

7,043

Customers

26.5%

Churn Rate

73.5%

Retention Rate

32.4 mo

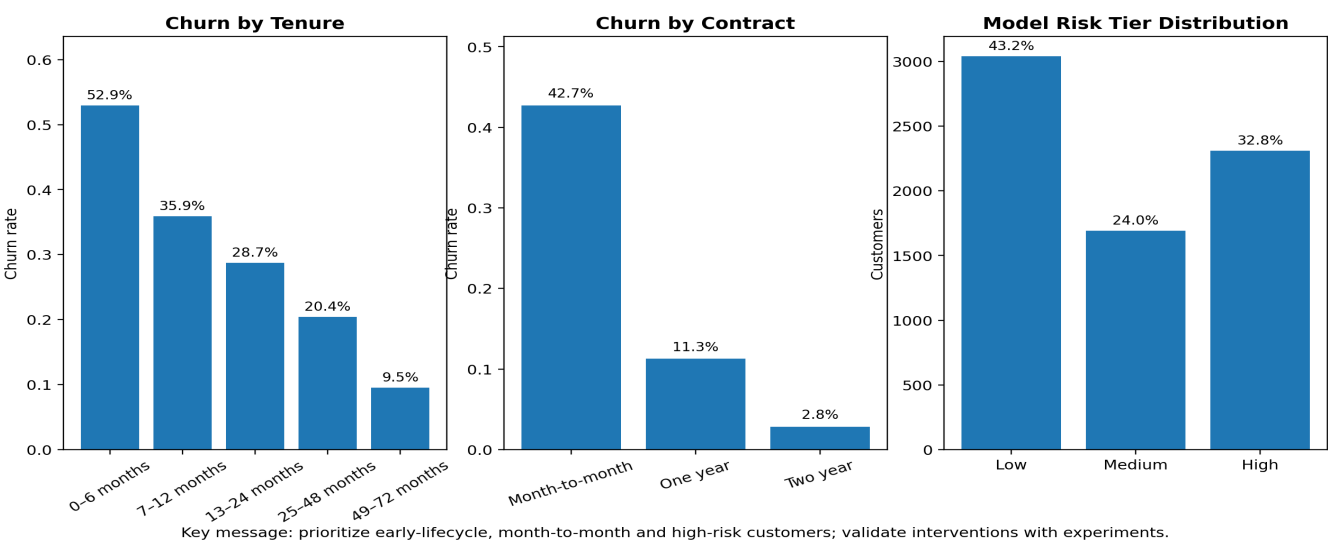
Avg Tenure

\$64.76

Avg Monthly Charges

0.845

Model ROC-AUC



Executive summary. The supplied dataset contains 7,043 customers, with 1,869 churned and 5,174 retained. Overall churn is 26.5%. The strongest observed churn concentration occurs early in the customer lifecycle, while month-to-month customers have materially higher churn than customers on longer contracts.

1. Executive KPIs

KPI	Result
Customers	7,043
Churn rate	26.5%
Retention rate	73.5%
Average tenure	32.4 months
Average monthly charges	\$64.76
Historical monthly charges associated with churned customers	\$139,130.85

2. Key findings

The churn rate declines from 52.9% for 0–6 month customers to 9.5% for 49–72 month customers. These are tenure-based lifecycle cohorts because the dataset does not provide signup dates.

Tenure band	Customers	Churn rate
0–6 months	1481	52.9%
7–12 months	705	35.9%
13–24 months	1024	28.7%
25–48 months	1594	20.4%
49–72 months	2239	9.5%

Contract type is strongly associated with churn: month-to-month 42.7%, one-year 11.3%, and two-year 2.8%. This is association, not causal proof.

Contract	Customers	Churn rate
Month-to-month	3875	42.7%
One year	1473	11.3%
Two year	1695	2.8%

3. Statistical testing

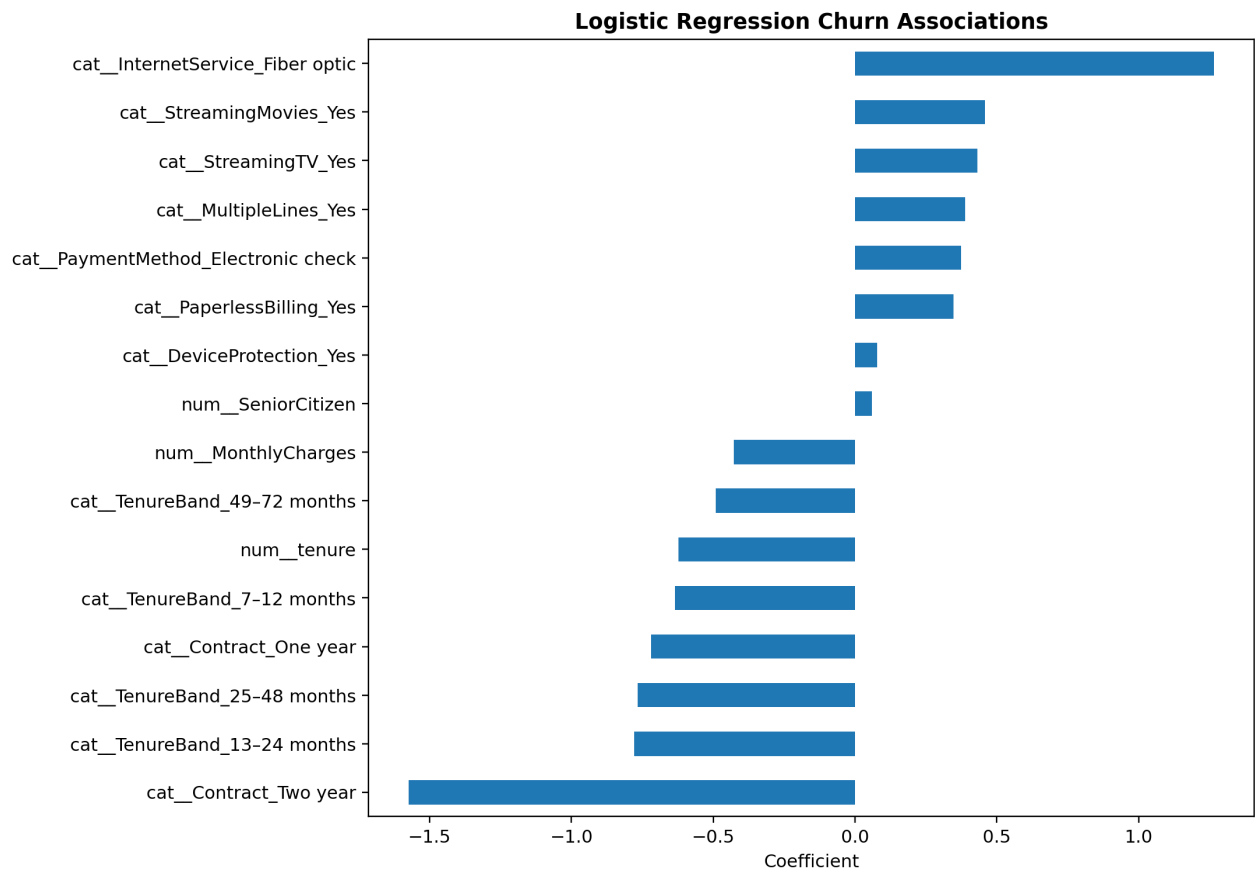
Chi-square tests were used for categorical variables and Welch's t-tests for tenure, monthly charges and total charges. Benjamini–Hochberg false-discovery-rate correction was added for the multiple categorical tests.

Feature	p-value	BH adjusted p	Significant after BH
Contract	5.86304e-258	8.20825e-257	True
OnlineSecurity	2.66115e-185	1.8628e-184	True
TechSupport	1.44308e-180	6.73439e-180	True
InternetService	9.57179e-160	3.35013e-159	True
PaymentMethod	3.68235e-140	1.03106e-139	True
OnlineBackup	2.07976e-131	4.85277e-131	True
DeviceProtection	5.50522e-122	1.10104e-121	True
StreamingMovies	2.66776e-82	4.66857e-82	True
StreamingTV	5.52899e-82	8.60066e-82	True
PaperlessBilling	4.07335e-58	5.7027e-58	True

4. Churn prediction

An interpretable Logistic Regression pipeline achieved 0.845 ROC-AUC and 0.805 recall on a stratified 20% holdout set. The model is evaluated out-of-sample, then refit on all historical records for final portfolio scoring.

Metric	Result
Accuracy	73.9%
Precision	50.8%
Recall	80.5%
F1	62.3%
ROC-AUC	0.845



5. Retention strategy

- **Early lifecycle:** strengthen onboarding, activation and first-value milestones during the first 6–12 months.
- **Risk-based outreach:** prioritize active high-risk/high-value customers for retention intervention.
- **Contract strategy:** test suitable month-to-month to longer-contract conversion offers.
- **Support/service:** investigate service and support gaps in high-churn segments.
- **Reason capture:** collect cancellation reasons, exit surveys, support tickets and satisfaction data.
- **Experimentation:** measure incremental retention with controlled experiments rather than treating association as causation.

6. Limitations & next steps

The supplied dataset has no signup date, region or explicit churn reason. It also lacks product-usage and engagement history. Therefore, the project does not claim true calendar-month cohorts, geographic churn reasons, or stated cancellation reasons. Future work should add these fields, build true cohort retention and CLV models, and validate the risk model on current/future business data.

Conclusion: the project provides a complete path from data quality to business action—lifecycle segmentation, statistical evidence, interpretable churn prediction, risk prioritization and retention strategy.